

Answer Key & Teacher Notes

Session 1 – Find the Whole • Session 2 – Solve for the Whole

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Find the Whole

TALK PROMPTS

- Why does finding 10% first make the whole easier to build?
- Before you scale up, how do you know the total has to be greater than the part?

PARTNER TASK

Puzzle Pieces and Water Bottle. One partner steps down to 10% (or 5%) on a double number line; the other builds up to 100% and checks forward with the percent. Switch roles with Library Books: 48 books is 60% (80 books).

WATCH FOR

- Multiplying the part by the percent (0.8×36) instead of scaling up, giving a whole smaller than the part.
- Dividing by the percent as a whole number ($36 \div 80$) instead of stepping down to 10%.
- Forgetting that the whole is 100% and stopping at the 10% amount.

SESSION 2 • TEACHER NOTES

Solve for the Whole

TALK PROMPTS

- Why does the unknown whole have to be larger than the part in a problem like this one?
- Why does dividing 312 by 0.65 give a number larger than 312?

PARTNER TASK

Fundraiser. A team raised \$312, which is 65% of its goal. One partner writes the equation $0.65 \times w = 312$; the other solves ($w = \$480$) and checks forward. Switch roles with Concert Ticket.

WATCH FOR

- Multiplying the part by the percent ($w = 0.25 \times 52$) instead of dividing.
- Dividing by the percent number ($1,050 \div 35$) instead of the decimal ($1,050 \div 0.35$).
- Putting the known amount in place of the whole in the equation ($w = 0.75 \times 27$).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1
S1 • Form B	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1
S2 • Form A	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

part: The amount you **know**; it matches a percent.

whole: The full amount. It is **100%**.

double number line: Two lines that **match** amounts with percents.

- The whole is 100%. When the percent is less than 100%, the whole is **greater** than the part.
- Divide to find a small percent like **10%**, then multiply up to 100%.
- Check: the percent of your whole gives back the **part**.

Example: $40\% \div 4 = 10\%$, so $24 \div 4 = 6$ stickers. $10\% \times 10 = 100\%$, so $6 \times 10 = 60$ stickers. Check: $0.40 \times 60 = 24$
Answer: The pack has **60** stickers.

YOU TRY

1. a. 80 b. 60 c. 50
2. $25\% = 6$; students: 6, 12, 18, 24. The class has 24 students.

EXIT TICKET

1. a. 110 b. 36
2. $10\% = 24 \div 4 = 6$; $100\% = 60$ games
3. **C** (greater than the part)

PRACTICE FORM A

1. a. 90 b. 80 c. 70 d. 48 e. 50 f. 40
2. a. \$18 ($54 \div 3$) b. \$180 (18×10). Check: $0.30 \times 180 = 54$.
3. a. $84 \div 7 = 12$ b. $12 \times 10 = 120$ pieces
4. a. $36 \div 3 = 12$ mL b. $12 \times 20 = 240$ mL
5. $10\% = 48 \div 6 = 8$ books; $100\% = 8 \times 10 = 80$ books
6. No. $10\% = \$21 \div 7 = \3 , so $100\% = \$30$. Kip multiplied 0.7×21 ; the first price must be more than \$21.
7. $25\% = 96 \div 3 = 32$; seats: 32, 64, 96, 128. There are 128 seats.
8. a. $4\% = 84 \div 7 = 12$; $100\% = 12 \times 25 = 300$ books b. $300 - 84 = 216$ books c. $0.28 \times 300 = 84$, which matches Friday
9. The first price must be more than the \$36 sale price, so multiplying by 0.8 is the wrong move. $10\% = 36 \div 8 = \$4.50$, so $100\% = \$45$. Check: $0.8 \times 45 = 36$.
10. **B** (60)
11. **C** (48)

Self-check word SCALING

PRACTICE FORM B

1. a. 70 b. 90 c. 84 d. 60 e. 80 f. 30
2. a. \$14 ($42 \div 3$) b. \$140 (14×10). Check: $0.30 \times 140 = 42$.
3. a. $63 \div 7 = 9$ b. $9 \times 10 = 90$ pieces
4. a. $27 \div 3 = 9$ mL b. $9 \times 20 = 180$ mL
5. $10\% = 42 \div 6 = 7$ books; $100\% = 7 \times 10 = 70$ books
6. No. $10\% = \$28 \div 7 = \4 , so $100\% = \$40$. Zara multiplied 0.7×28 ; the first price must be more than \$28.
7. $25\% = 108 \div 3 = 36$; seats: 36, 72, 108, 144. There are 144 seats.
8. a. $4\% = 72 \div 6 = 12$; $100\% = 12 \times 25 = 300$ books b. $300 - 72 = 228$ books c. $0.24 \times 300 = 72$, which matches Monday
9. The first price must be more than the \$45 sale price, so multiplying by 0.9 is the wrong move. $10\% = 45 \div 9 = \$5$, so $100\% = \$50$. Check: $0.9 \times 50 = 45$.
10. **B** (50)
11. **C** (50)

Self-check word PERCENT

GUIDED NOTES

variable: A letter that stands for an **unknown** number.

equation: A math sentence with an **equal** sign.

inverse operation: An operation that **undoes** another.
Division undoes multiplication.

- Write the equation percent $\times$ whole = part, with the percent as a **decimal**.
- Divide the part by the decimal to find **w**.
- Check: multiply your **w** by the percent to get the **part**.

Example: $35\% = 0.35$ Equation: $0.35 \times w = 42$ $w = 42 \div 0.35 = 120$ Answer: The team scored **120** points.

YOU TRY

1. a. 46 b. 45 c. 50
2. $0.15 \times w = 7.20$; $w = 7.20 \div 0.15 = \$48$

EXIT TICKET

1. a. 52 b. 70
2. $0.6 \times w = 138$; $w = 138 \div 0.6 = \$230$
3. **B** (40)

PRACTICE FORM A

1. a. 80 b. 60 c. 76 d. 68 e. 70 f. 65
2. a. $0.7 \times w = 18,900$ b. $w = 18,900 \div 0.7 = 27,000$; they paid \$27,000
3. $0.4 \times w = 84$; $w = 84 \div 0.4 = \$210$
4. a. $0.8 \times w = 96$, so $w = 120$ minutes b. $120 - 96 = 24$ minutes
5. $0.65 \times w = 312$; $w = 312 \div 0.65 = \$480$
6. No. The unknown whole goes with the percent: $0.75 \times w = 27$, so $w = 27 \div 0.75 = \$36$.
7. $0.35 \times w = 1,050$; $w = \$3,000$. Left after rent: $\$3,000 - \$1,050 = \$1,950$.
8. a. $0.7 \times j = 63$, $j = \$90$; $0.45 \times b = 63$, $b = \$140$ b. the boots: $\$140 - \$63 = \$77$ off, compared with $\$90 - \$63 = \$27$ off c. \$63 is a smaller percent of the boots' price, so the boots' first price had to be greater
9. Her whole came out smaller than the part. \$52 is the part, so the equation is $0.25 \times w = 52$, and $w = 52 \div 0.25 = \$208$. Check: $0.25 \times 208 = 52$.
10. **B** ($0.2 \times w = 30$)
11. **C** (60)

Self-check word **DIVIDE**

PRACTICE FORM B

1. a. 70 b. 48 c. 68 d. 82 e. 60 f. 55
2. a. $0.6 \times w = 16,800$ b. $w = 16,800 \div 0.6 = 28,000$; they paid \$28,000
3. $0.4 \times w = 66$; $w = 66 \div 0.4 = \$165$
4. a. $0.8 \times w = 72$, so $w = 90$ minutes b. $90 - 72 = 18$ minutes
5. $0.65 \times w = 273$; $w = 273 \div 0.65 = \$420$
6. No. The unknown whole goes with the percent: $0.75 \times w = 24$, so $w = 24 \div 0.75 = \$32$.
7. $0.35 \times w = 1,120$; $w = \$3,200$. Left after rent: $\$3,200 - \$1,120 = \$2,080$.
8. a. $0.8 \times j = 72$, $j = \$90$; $0.45 \times b = 72$, $b = \$160$ b. the boots: $\$160 - \$72 = \$88$ off, compared with $\$90 - \$72 = \$18$ off c. \$72 is a smaller percent of the boots' price, so the boots' first price had to be greater
9. She multiplied instead of dividing, so her whole is smaller than the part. $w = 70 \div 0.35 = 200$. Check: $0.35 \times 200 = 70$.
10. **B** ($0.3 \times w = 18$)
11. **C** (60)

Self-check word **EQUALS**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · FIND THE WHOLE

1. $20\% = \$18$; dollars: 18, 36, 54, 72, 90. The helmet costs \$90.
2. a. 130 b. 44 c. 70 d. 80 e. 140 f. 90
3. One part (25%) = $18 \div 3 = 6$ gal; full = $4 \times 6 = 24$ gallons.
4. $25\% = 120 \div 3 = 40$ pages, so $100\% = 4 \times 40 = 160$ pages.
5. A is greater: 20 is a smaller percent (40%) of A, so A must be bigger. $A = 50$ ($10\% = 5$) and $B = 25$ ($10\% = 2.5$, or $20 \div 0.8$).

SHEET 2 · SOLVE FOR THE WHOLE

1. a. 70 b. 48 c. 80
2. a. $0.55 \times w = 33$ b. $w = 33 \div 0.55 = 60$
3. $0.75 \times w = 315$, $w = 315 \div 0.75 = \$420$. Still needed: $\$420 - \$315 = \$105$.
4. $0.35 \times w = 42$, $w = 42 \div 0.35 = 4,200 \div 35 = \120 . Check: $0.35 \times 120 = 42$.
5. He divided by 35 instead of 0.35 ($35\% = 0.35$); a whole of 0.8 is smaller than the part, which makes no sense. $w = 28 \div 0.35 = 80$. Check: $0.35 \times 80 = 28$.
6. **B** ($0.3 \times w = 45$)
7. **C** (30)